Design and Implementation of a Modular Dual-Evaporator Transcritical High Temperature Heat Pump

Research Project

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1 Context

According to the Global Energy Review 2025 report published in March 2025 by the IEA [2], global CO_2 emissions increased by 0.8% in 2024, reaching an all-time high of 37.8 Gt CO_2 per year. In addition, the report highlights that energy demand grew by 2.2% in 2024. This increase in CO_2 emissions is inconsistent with the goals of the Paris Agreement, which aims to reduce CO_2 emissions by 43% by 2030 and to reach net-zero emissions by 2050. However, the lower increase in CO_2 emissions compared to energy demand growth is an encouraging trend, as it suggests a relative decoupling of emissions from energy consumption.

Kosmadakis et al. report that the waste heat potential in the EU-27 and the UK reached 221.32 TWh/year in 2021 [5]. In comparison, the demand for heat below 140°C in Germany was 612 PJ, or 170 TWh, in 2012 [4]. These numbers justify the interest of companies and researchers in designing and studying machines capable of recovering this waste energy.

In this work, we will focus on the use of high-temperature heat pumps (HTHPs) for recovering waste heat from industrial processes. Arpagaus et al. provide a list of manufacturers and universities that have already been working on the design of several machines for at least one decade [4]. The existing machines cover a large spectrum of applications, ranging from small units of a few kilowatts to large ones of more than one megawatt. However, new design ideas can still be studied in order to improve current recovery systems.

This thesis focuses on the design, construction, and performance evaluation of a modular machine capable of operating under various thermodynamic cycles. Among these, the dual-evaporator transcritical cycle, whose schematic is shown in Figure 5, represents the most elaborate and complex configuration implemented. A review of the existing literature did not reveal any previous applications of such a dual-evaporator transcritical cycle for HTHP systems. Furthermore, the machine developed within the framework of this master's thesis is intended to serve as a platform for future academic and research activities.

2 Literature review

As discussed in the previous section, no publications specifically addressing dual-evaporator transcritical HTHPs were identified. Nevertheless, a brief review of the current state-of-the-art technology for HTHPs was carried out. This review has led to the identification of several reasons why the topic may be particularly relevant for certain applications and therefore is worth further investigation.

First, regarding the dual-evaporator configuration, Mateu-Royo et al. show that using multiple evaporators offers significant potential for waste heat recovery, as it enables simultaneous recovery of heat from sources at different temperature levels [6]. In the case study considered, notable improvements in both the Coefficient of Performance (COP) and Volumetric Heating Capacity (VHC)[1] were observed when two evapora-

[1] The Volumetric Heating Capacity (VHC), expressed in kJ/m^3, is the ratio of the thermal power delivered at the heat sink to the volumetric flow rate of the working fluid.

tors were employed instead of one, once a significant portion of the available heat was present at the medium temperature (MT) level (see Figure 1).

This proportion of heat available at the medium temperature level relative to the total heat input is quantified by the heat source ratio parameter (β), first introduced by Arpagaus et al. [3]:

$$\beta \triangleq \frac{\dot{Q}_{MT}}{\dot{Q}_{MT} + \dot{Q}_{LT}}$$

where $\dot{Q}_{MT}$ and $\dot{Q}_{LT}$ represent the thermal powers captured at the medium and low temperature evaporators, respectively.

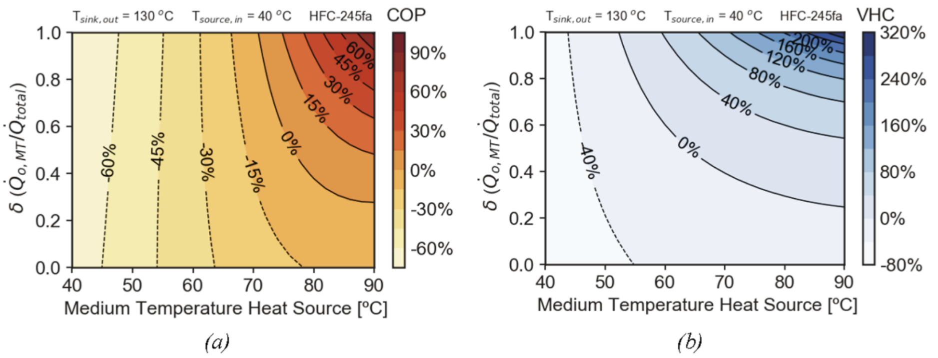

Fig. 15. Performance parameters evaluation of the TS Booster cycle compared to the reference TS Economizer, operating with HFC-245fa: a) COP and b) VHC.

Figure 1: COP and VHC increase significantly when using two evaporators instead of one, once a substantial portion of the total captured heat is available at the medium-temperature source. Figure taken from [6].

While the benefit of a higher COP is not to be demonstrated, the VHC is also of particular interest. As explained by Arpagaus et al., the VHC is directly related to the refrigerant charge and size of the compressor: a higher VHC allows for a smaller refrigerant charge and a more compact compressor, thereby reducing system costs [4]. Consequently, both the COP and the VHC are important factors in designing efficient and cost-effective heat pumps.

Next, regarding the transcritical aspect of the cycle: Vieren et al. demonstrate that using a transcritical cycle instead of a subcritical one can lead to significant improvements in COP when the temperature glide at the heat sink is high [7]. For instance, a COP improvement of 4.6% over the subcritical cycle was reported for a heat sink temperature glide of 81 K. According to their study, the ideal heat sink temperature glide is around 60 K, taking into account both practical and economic considerations. This COP increase is attributed to a lower temperature mismatch during heat transfer at the heat sink (see Figure 2). The improved temperature matching results from the properties of a supercritical fluid. Since heat transfer occurs above the critical pressure, the fluid does not undergo a phase change. Therefore, its temperature can vary significantly during heat transfer, allowing for a better match with the heat sink fluid temperature profile and reduced exergy losses.

Vieren et al. also show that transcritical cycles consistently achieve higher VHC values than subcritical ones. For example, at a heat sink temperature glide of 81 K, the best transcritical cycle (with R1336mzz(Z)) nearly doubles the VHC of the best subcritical case (with ethanol), although the magnitude of improvement varies with the working fluid and operating conditions (see Tables 5-7 in [7]).

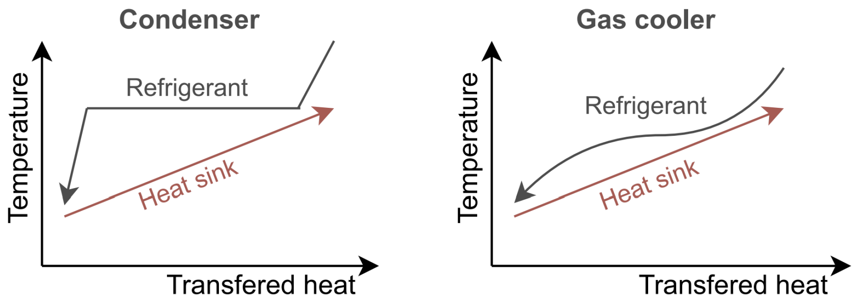

Fig. 1. Temperature profile during heat rejection for subcritical operation (condenser) and transcritical or supercritical operation (gas cooler).

Figure 2: The area between the two curves represents the exergy destruction due to heat transfer. This area is considerably smaller in the transcritical case, resulting in lower irreversibilities and a potentially higher COP. Figure taken from [7].

Finally, regarding high-temperature applications: heat pumps are classified as "high temperature" when the heat sink temperature exceeds 100 °C [4]. While conventional heat pumps are already widely used for applications below 100 °C, HTHPs based on vapor-compression cycles (i.e., inverse Rankine cycles) are designed to provide process heat in the range of 100-200°C. According to the IEA, nearly 20% of industrial heat demand falls within the temperature range suitable for these heat pumps (see Figure 3) [1].

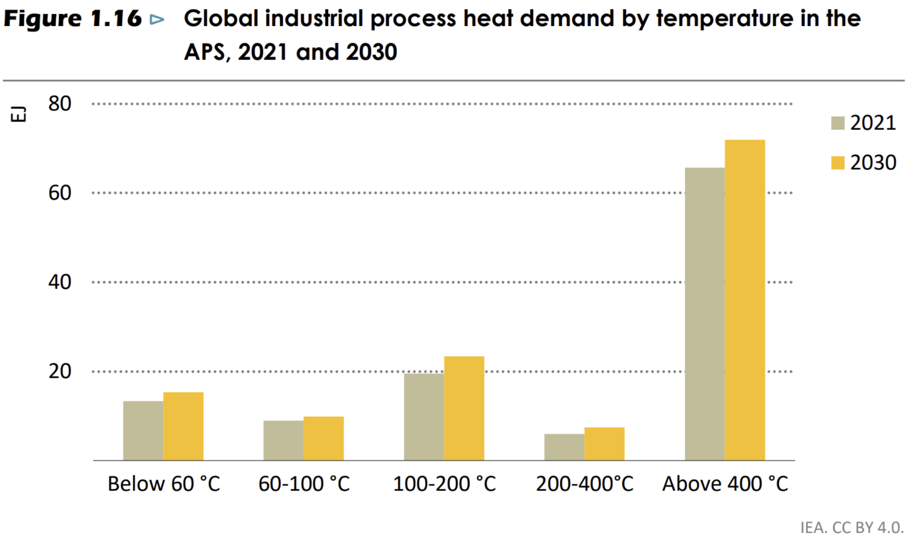

Figure 3: Around 22 EJ of industrial heat demand will be required in the range of 100-200 °C, corresponding to just under 20% of the expected total industrial heat demand in 2030. Figure taken from [1].

Today, most of this industrial heat is based on fossil fuels. Developing efficient high-temperature heat pumps is therefore a crucial step toward decarbonizing this 22 EJ of industrial heat demand between 100°C and 200°C, and more broadly, supporting the energy transition.

In conclusion, a dual-evaporator transcritical high-temperature heat pump has the potential to outperform current HTHPs on the market, particularly in applications where heat is available at two temperature levels (e.g., waste heat recovery) and where the temperature glide at the heat sink is high. The aim of this master's thesis is to investigate this concept, contributing to the broader effort to advance high-temperature heat pumps and decarbonize industrial heat production.

3 Research question

While refrigeration cycles have been the subject of extensive efficiency studies in recent years, HTHP cycles have received comparatively little attention. Our objective is to implement new design ideas inspired by these studies, such as the dual evaporator or the transcritical cycle, to improve the efficiency of HTHP cycles.

This work is addressed to the scientific community, but also to industry, since it aims to provide insights and solutions that can be directly applied in real-world applications. In addition, a prototype will be developed to validate the proposed concepts.

Our work is also multidisciplinary, as it includes thermodynamics, heat transfer, fluid mechanics, control techniques, and machine design. It is a complete project, from the numerical design to the building and testing of a prototype.

The research question that will be addressed in this master's thesis is the following:

How can a modular high-temperature heat pump be designed, sized, controlled, and prototyped to operate in both subcritical and transcritical modes, with optional use of recuperators and the flexibility to employ one or two evaporators?

4 Project description

To address the previous research question, this master's thesis is structured into eight steps, as shown in Figure 4.

Preliminary Thermodynamic Program

In this first part, a classical heat pump cycle will be modeled numerically. Each student will work simultaneously on different components, such as a semi-empirical model for the compressor or a moving-boundary model for the heat exchangers. The cycle analysis will include both energy and exergy analyses to ensure optimal performance. The numerical model will be developed in `Python`, using the `CoolProp` library to access thermodynamic property data.

Final Thermodynamic Program

First, the various cycle configurations will be optimized to identify the most suitable components for each. Then, the program will be generalized to enhance its robustness and flexibility.

Sizing of the Main Components and P&ID

After identifying the optimal components for each configuration, a trade-off analysis will be carried out to select the final components of the machine. Simulations will then be performed to verify that the system can operate in all configurations using the selected components. In addition, a P&ID (Piping and Instrumentation Diagram) will be developed to illustrate the system layout.

Establishment of a First Control Strategy

One of the main advantages of this machine lies in its ability to simulate different cycle configurations. Therefore, a suitable control logic must be implemented. This step is crucial for determining the appropriate type and placement of sensors.

CAD and Operational Model

In this section, the mechanical design of the machine will be carried out. This includes material and component selection, the design of heat exchangers and insulation, and the creation of detailed technical drawings. The resulting operational model will also serve to analyze the machine's dynamic behavior.

Prototype Construction and Instrumentation

This stage involves constructing the prototype and installing the necessary instrumentation.

Safety Validation of the Prototype

Before testing, the prototype will be validated by the CREDEM platform to ensure compliance with safety requirements.

Control Logic and Performance Validation

Finally, the last stage will focus on evaluating the machine and its control logic. A series of tests will be conducted to verify proper operation and to collect data for comparison with theoretical predictions.

5 Share of workload and responsibilities

Taking the different steps of the project presented in the previous section, the workload will be shared as follows:

1. **Preliminary Thermodynamic Program:** The EPL students will work in parallel on different sub-parts of the `Python` code, but strategic design decisions will be made jointly and mutual support will be provided as needed.

2. **Final Thermodynamic Program:** The optimization stategy for each configuration will be jointly developed by the EPL students to maximize the performance of the system in nominal conditions.

3. **Sizing of the Main Components and P&ID:** Similar to the first phase, the EPL students will work in parallel on different components to maximize productivity, as this phase must be completed by the end of the first semester to ensure availability of the components for the prototyping phase.

4. **Establishment of a First Control Strategy:** All students will participate in the development of a first control strategy for the prototype.

5. **CAD and Operational Model:** The ECAM student will lead the CAD work, with technical support from a CREDEM platform technician. The resulting Operational Model will be validated by the EPL students based on their initial thermodynamic design.

6. **Prototype Construction and Instrumentation:** The prototype will be built under the supervision of a CREDEM technician.

7. **Safety Validation of the Prototype:** This step will be fully managed by the CREDEM platform.

8. **Control Logic and Performance Validation:** Once the prototype is operational, the students will carry out the validation of control logic and performance, as well as fine-tune the prototype to optimize its operation.

Throughout the project, weekly meetings will be organized between the students to discuss progress, share knowledge, and make important decisions. In addition, students will meet regularly with their mentors to present key results, receive feedback, and discuss challenges. Progress will also be reported formally during monthly meetings with the supervisor.

6 Project team

The project team is composed of the following members:

- **EPL students:** Warscotte Hugo, Grégoire Franklin
- **ECAM student:** To be determined
- **CREDEM platform technician:** Hesbois Frank
- **Mentors:** Hauglustaine Mattéo, Laterre Antoine
- **Supervisor:** Prof. Contino Francesco

7 Roadmap and planning

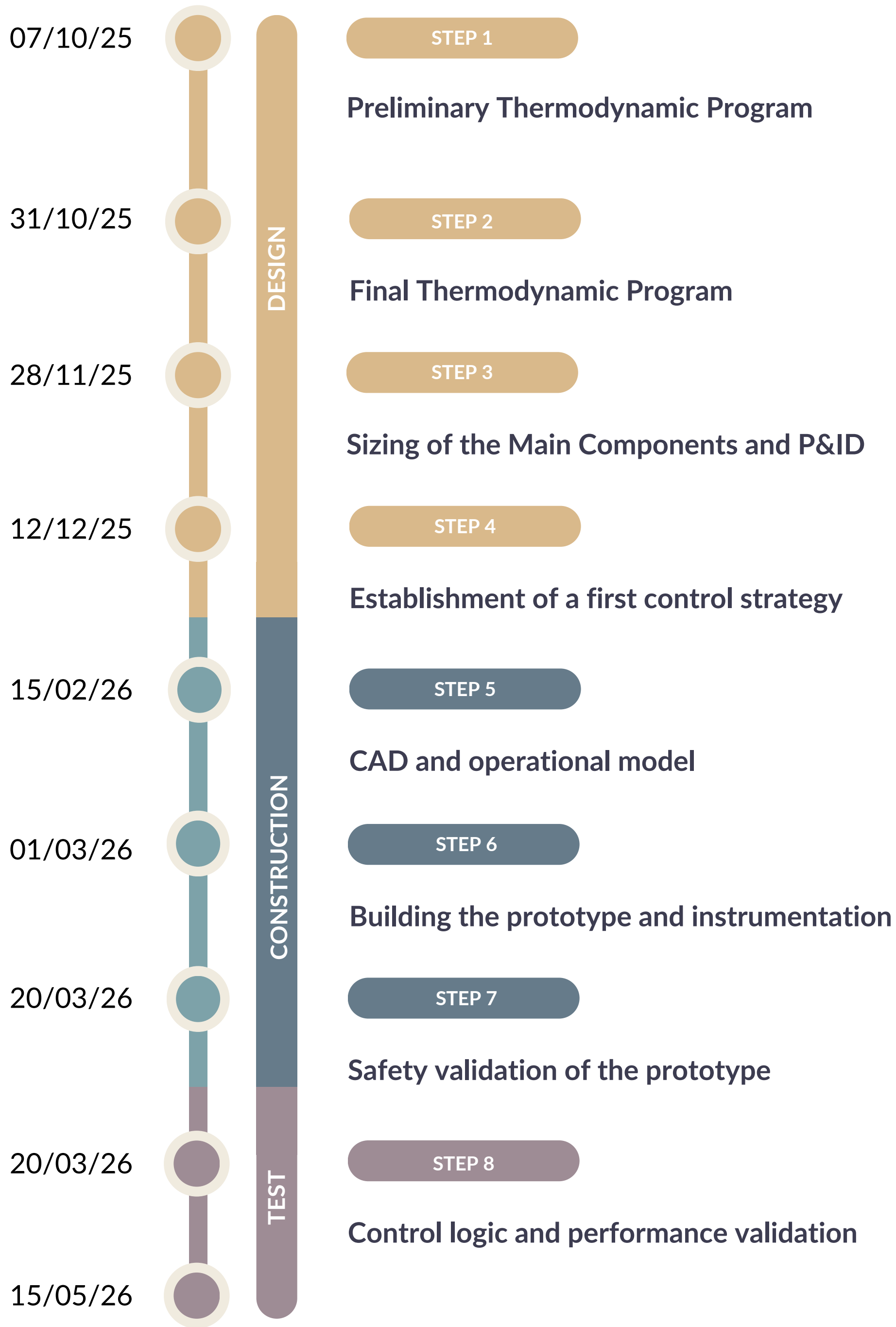

Figure 4: This figure shows the different steps of the master's thesis on the right, with the corresponding dates on the left.

References

[1] International Energy Agency. The future of heat pumps. `https://www.iea.org/reports/the-future-of-heat-pumps`, 2022. Accessed: 2025-09-27.

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[7] Elias Vieren, Toon Demeester, Wim Beyne, Alessia Arteconi, Michel De Paepe, and Steven Lecompte. The thermodynamic potential of high-temperature transcritical heat pump cycles for industrial processes with large temperature glides. *Applied Thermal Engineering*, 234:121197, 2023.

Note on AI assistance

This report was proofread with the assistance of `ChatGPT`, an AI language model developed by `OpenAI`.

Appendix

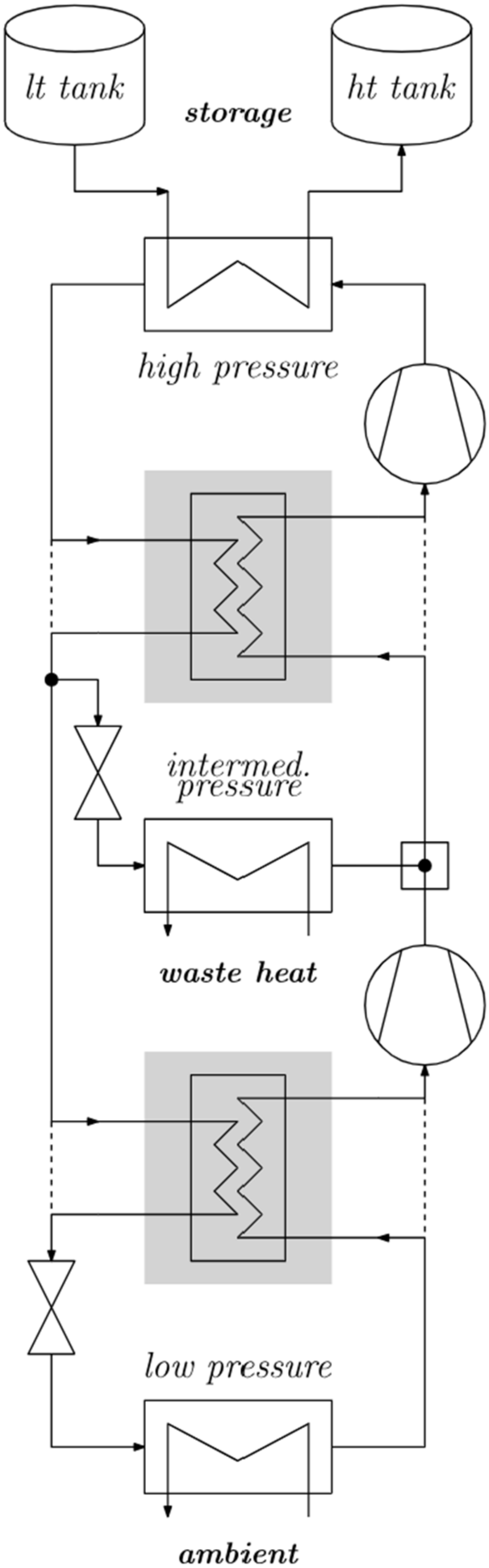

Figure 5: Schematic of the thermodynamic cycle of a dual-evaporator transcritical high temperature heat pump.